23/10/25

Bivariate jointly Gaussian Distribution

"X, Y: bivariate jointly Gaussian if

$aX+bY \sim$ Normal for all a, b.

$X \sim N(\mu_x, \sigma_x^2)$ $\quad Y \sim N(\mu_y, \sigma_y^2)$ How are $X+Y$ correlated?

$U, V \sim$ iid $N(0,1)$

$\mu_x = \mu_y = 0$

$\sigma_x = \sigma_y = 1$

$X = U$

$Y = \rho U + \sqrt{1-\rho^2}\, V$

$E[XY] = \rho$ "correlation coefficient"

$X_1 = \sigma_x X + \mu_x, \quad X_2 = \sigma_y Y + \mu_y$

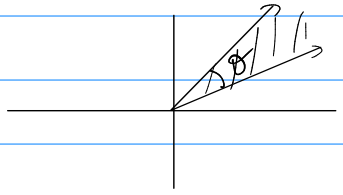
$Cov(X,Y) = E[(X-\mu_x)(Y-\mu_y)]$

$$f_{XY}(x,y) = \frac{1}{2\pi\sqrt{1-\rho^2}}\, e^{-\left(\frac{x^2-2\rho xy+y^2}{2(1-\rho^2)}\right)} \quad \text{all } x, y.$$

$$f_{X_1X_2}(x_1,x_2) = \frac{1}{2\pi\sigma_x\sigma_y\sqrt{1-\rho^2}}\, e^{-\left(\left(\frac{x_1-\mu_x}{\sigma_x}\right)^2 - 2\rho\left(\frac{x_1-\mu_x}{\sigma_y}\right)\left(\frac{x_2-\mu_y}{\sigma_y}\right)+\left(\frac{x_2-\mu_y}{\sigma_y}\right)^2\right)}{2(1-\rho^2)} \quad \text{all } x_1, x_2$$

① $X, Y \sim$ iid $N(0,1)$ $\quad Pr(X > Y) = 1/2$ $\quad Pr(X>0, Y>0) = 1/4$

$R = \sqrt{X^2+Y^2}$ $\quad \Theta = \tan^{-1} Y/X \sim$ Uniform $[0, 2\pi]$ (4-quadrant $\tan^{-1}$)



② $X, Y \sim$ Bivariate Gaussian $\quad \mu_x = \mu_y = 0$, $\sigma_x = \sigma_y = 1$, ρ

$Pr(X > Y) = 1/2$

$$Pr(X>0, Y>0) = \int_{x=0}^{\infty}\int_{y=0}^{\infty} \frac{1}{2\pi\sqrt{1-\rho^2}}\, e^{-\left(\frac{x^2-2\rho xy+y^2}{2(1-\rho^2)}\right)} dx\, dy$$

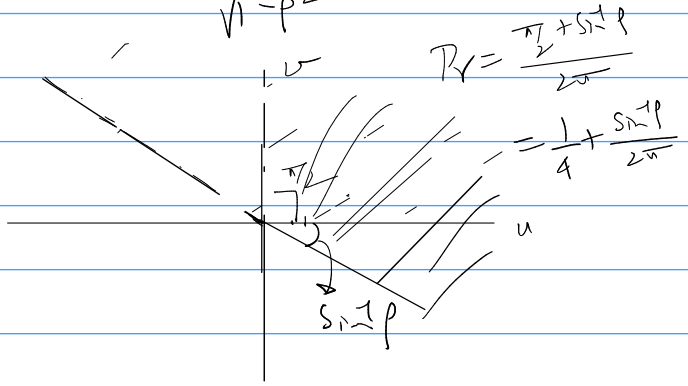
$U, V \sim \text{iid } N(0,1)$

$$X = U$$

$$Y = \rho U + \sqrt{1-\rho^2}\, V$$

$$X > 0,\ Y > 0 \iff U > 0$$

$$V > \frac{-\rho}{\sqrt{1-\rho^2}} U$$



$$P_r = \frac{\frac{\pi}{2} + \sin^{-1}\rho}{2\pi} = \frac{1}{4} + \frac{\sin^{-1}\rho}{2\pi}$$

Conditional densities $U, V \sim \text{iid } N(0,1)$

$$X = U$$

$$Y = \rho U + \sqrt{1-\rho^2}\, V$$

$\mu_X = \mu_Y = 0$

$\sigma_X = \sigma_Y = 1$

$$Y \mid X = x \sim N(\rho x,\ 1-\rho^2)$$

$$E[Y \mid X = x] = \rho x$$

$$E[Y \mid X] = \rho X$$

More general

$$\frac{X - \mu_X}{\sigma_X} = U$$

$$\frac{Y - \mu_Y}{\sigma_Y} = \rho U + \sqrt{1-\rho^2}\, V$$

$$\frac{Y-\mu_Y}{\sigma_Y} \mid X = x \sim N\left(\rho\frac{(x-\mu_X)}{\sigma_X},\ 1-\rho^2\right)$$

$$Y \mid X = x \sim N\left(\mu_Y + \frac{\rho\sigma_Y(x-\mu_X)}{\sigma_X},\ \sigma_Y^2(1-\rho^2)\right)$$

$$E[Y \mid X] = \frac{\rho\sigma_Y}{\sigma_X} X + \mu_Y - \frac{\rho\mu_X\sigma_Y}{\sigma_X}$$

$U_1, U_2, \ldots, U_n \sim \text{iid } N(0,1)$

$$X_1 = a_{11} U_1 + \cdots + a_{1n} U_n$$

$$\vdots$$

$$X_n = a_{n1} U_1 + \cdots + a_{nn} U_n$$

$$X = \begin{bmatrix} X_1 \\ \vdots \\ X_n \end{bmatrix} \qquad U = \begin{bmatrix} U_1 \\ \vdots \\ U_n \end{bmatrix} \qquad X = AU$$

$$A = [a_{ij}]_{1 \le i,j \le n}$$